

6 A photocell is connected in a series circuit with a variable d.c. power supply and a sensitive ammeter as shown in Fig. 6.1.

The photocell is illuminated with electromagnetic radiation of wavelength 264 nm and power 3.8 mW and photoelectrons are emitted. The potential difference V between the collector C and emitter E in the photocell is adjusted and the photocurrent I is measured. Fig. 6.2 below shows the graph of I against V .

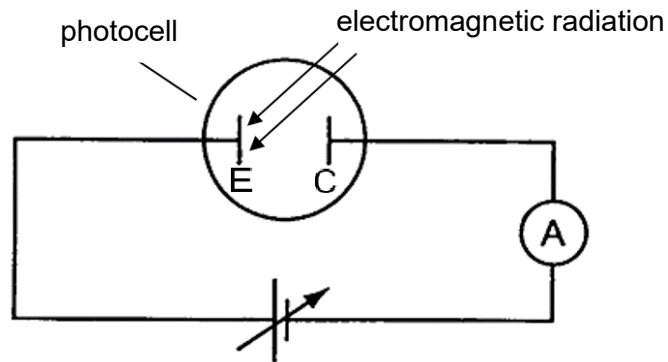


Fig. 6.1

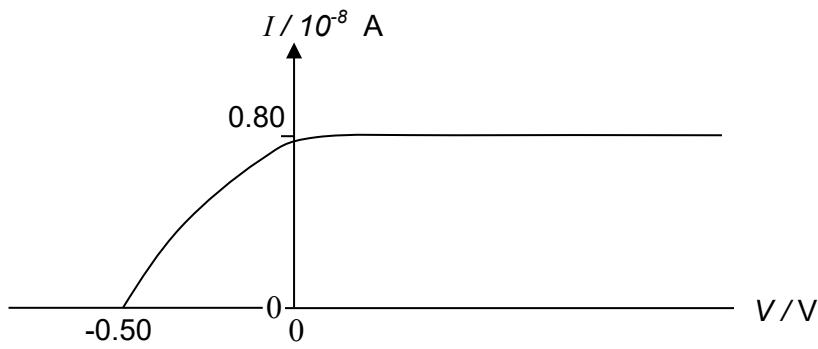


Fig. 6.2

(a) Explain why the photocurrent does not continue to increase for positive values of V .

.....

..... **[1]**

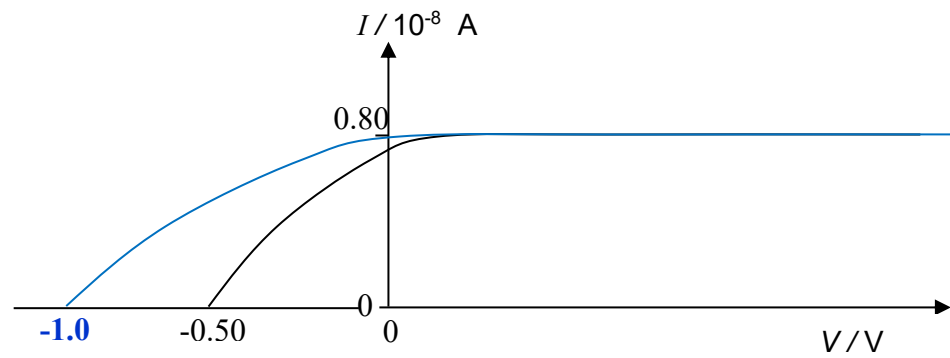
Solution:

The rate of electron ejection from the emitter arriving at the collector is fixed because it is proportional to the rate of photon incidence which is fixed when electromagnetic radiation is at a fixed intensity and fixed wavelength.

	(b)	(i)	Determine the energy of a photon of the incident electromagnetic radiation.	
			energy = J	[1]
			Solution: <i>Energy of a photon</i> = $\frac{hc}{\lambda}$ $= \frac{6.63 \times 10^{-34} (3.00 \times 10^8)}{(264 \times 10^{-9})}$ $= 7.5341 \times 10^{-19} \text{ J}$ $= 7.53 \times 10^{-19} \text{ J}$	1
		(ii)	Calculate the rate at which the photons are incident on the emitter.	[2]
			<i>Rate of photon incidence</i> = $\frac{\text{incident power}}{\text{energy of photon}}$ $= \frac{3.8 \times 10^{-3}}{7.5341 \times 10^{-19}}$ $= 5.0437 \times 10^{15} \text{ s}^{-1}$ $= 5.0 \times 10^{15} \text{ s}^{-1}$	M1 A1
			rate of photon incidence = s ⁻¹	[2]
			Solution: <i>Rate of photon incidence</i> = $\frac{\text{incident power}}{\text{energy of photon}}$ $= \frac{3.8 \times 10^{-3}}{7.5341 \times 10^{-19}}$ $= 5.0437 \times 10^{15} \text{ s}^{-1}$ $= 5.0 \times 10^{15} \text{ s}^{-1}$	

		(iii)	Show that the maximum rate of photoelectron emission is $5.0 \times 10^{10} \text{ s}^{-1}$.	
				[1]
			Solution: <i>Rate of photoelectron emission</i> = $\frac{\text{photoelectric saturation current}}{\text{charge of electron}}$ $= \frac{0.80 \times 10^{-8}}{1.60 \times 10^{-19}}$ $= 5.0 \times 10^{10} \text{ s}^{-1}$	
		(iv)	Suggest a reason for the difference between (b)(ii) and your answer in (b)(iii).	
				
				[1]
			A fraction of the incident photons are reflected by the emitter and do not cause photoelectric effect, hence the number of emitted photoelectrons is smaller than the number of incident photons.	1
		(c) (i)	Calculate the work function energy of the emitter.	
			work function energy = J	[1]
			Solution: Using photoelectric equation, $E_{\text{photon}} = \phi + eV_s$ $7.5341 \times 10^{-19} = \phi + (1.60 \times 10^{-19})(0.50)$ $\phi = 6.7 \times 10^{-19} \text{ J}$	1

- (ii) The emitter is replaced with another emitter of **half** the work function energy. On Fig. 6.2, sketch a graph to show the new variation with V of I . [2]



1 mark: correct graph with a more negative horizontal intercept
1 mark: precision of horizontal intercept (doubled), labelling the new horizontal intercept at -1.0 V

7 A solar cell is a device which converts light energy directly into electrical energy. Under the influence